

Matthew Hwang

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Education

University of California, San Diego: La Jolla, CA

September 2023 - June 2027

Bachelor's of Science in *Mechanical Engineering*

UC-GPA: 3.576

Experience

Mechanical Design Intern | Nano Bioelectronics Laboratory: Dr. Joseph Wang

July 2025 - Present

- Prototyping a modular housing fixture to hold acupuncture needles in place for PDMS mold creation
- Designing a resin-printed circuit board & battery capsule in SolidWorks for a microneedle wearable sensor

Co-President | TUAS (Triton Unmanned Aerial Systems)

September 2024 - Present

- Managing school-affiliated and independent savings accounts ensuring accurate tracking of funds
- Rewrote financial documentation and club workflow providing efficient tracking sheets.
- Processing all purchase orders and ensuring clean spreadsheet documentation of expenditures
- Directed sub-teams across engineering and business in the design of the competition aircraft

Project Team Co-Chair | BMES (Biomedical Engineering Society)

July 2025 - Present

- Leading efforts to design & prototype an 8-DOF robotic arm controlled by a haptic glove worn by a user
 - Creating a haptic glove with spatial tracking system achieved with IMU sensors and magnetic encoders
 - Simulating sensory data with haptic glove via vibrotactile feedback using sensors embedded in robotic arm
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Projects

Autonomous Targeting System

September 2025 - December 2025

- Developed an Autonomous Targeting System using OpenCV with scan, track, and fire modes
- Implemented a 2-DOF 3D-printed gimbal and "Smart-Cam" to center targets and reacquire moving targets after losing them in frame
- Programmed an algorithm that detects objects and saves their positions in space and allows for manual selection of an object to be a target
- Designed an ignition system to fire a rocket engine at target and a sub-program to verify target was hit

Intro to Autonomous Vehicles | MAE/ECE 148

September 2025 - December 2025

- Collaborated with a team of 4 to develop an autonomous car with a DonkeyCar AI framework using RPI5
- Utilized a ROS2 container to run a CV program keeping a car between dotted and solid lane lines on track
- Integrated a GPS module and tuned PID parameters to accurately follow manually recorded driving paths

"Smart" Walker

September 2025

- Prototyped a low-cost sensor-integrated walker to amplify spatial awareness for visually impaired users
- Developed an Arduino-based multimodal feedback system to provide real-time visual and auditory alerts
- Integrated OpenCV with ESP32-CAM to identify obstacle position and proximity relative to user

CPR Training Kit

July 2025

- Co-designed a low-cost CPR training kit with a team of 3, aligning with AHA compression guidelines
- Programmed training system with CircuitPython, guiding users through chest compressions at 100 BPM
- Achieved reliable force detection from 0-100 N and a LED feedback system to rate accuracy

Mechanical Piston Heart Project | BMES (Biomedical Engineering Society)

September 2024 - May 2025

- Designed a slider-crank linkage using Fusion360 to convert rotary motion into linear piston actuation
- Calculate piston diameter and stroke length based on stroke speed and force parameters
- Collaborated with other subteams to ensure compatibility and smooth integration of the overall model

Mechanical Design Robot Contest | MAE 3

October 2024 - December 2024

- Directed a team of four in building a functional robot in six weeks, meeting all competition objectives
 - Performed risk reduction assessments, force/torque/energy/power evaluations to predict performance
 - Utilized Fusion360 and AutoCAD to design and fabricate robot parts with laser cutters and 3D printers
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Skills

Fusion360 | AutoCAD | SolidWorks | 3D Printing | Excel | Arduino | Python | Linux | Robot Operating System

